Operating Systems - Unit-wise Question Bank

CBIT previous papers (2021-2026), sorted by the R22 syllabus. Out-of-syllabus questions removed; identical questions merged and marked as repeated.

Papers used

- Dec 2024 (CSE/CET V)
- Nov 2025 (CSE/CET V)
- May 2025 (CSM/AIML IV)
- Jun 2024 (CSM/AIML IV)
- Jun 2024 (AIML VI)
- Jan 2023 (CSE V)
- Jan 2023 (IT/AIDS V)
- Dec 2021 (CSE V)
- Dec 2021 (IT V)
- Jun 2024 RTOS (ECE)
- May 2025 RTOS (ECE)
- May 2026 RTOS (ECE)

Unit	Short	Long	Problems	Total
UNIT I - Introduction to Operating Systems & OS Architectures	13	31	0	44
UNIT II - Process Management, Scheduling & IPC	13	31	7	51
UNIT III - Memory Management & Secondary Storage	11	23	15	49
UNIT IV - Concurrency, Synchronization & Deadlocks	10	31	4	45
UNIT V - File Systems & OS Security	12	29	0	41

UNIT I - Introduction to Operating Systems & OS Architectures

Syllabus: Computer system overview, components, functions of OS, types of OS (RTOS / desktop / mobile), distributions, micro-kernel, layered and kernel approaches, Linux/Unix design.

Short-answer questions (Part A)

1. What is the main difficulty that a programmer must overcome in writing an operating system for a real-time environment?
   [2M] Dec 2024 (CSE/CET V)
2. Describe the micro-kernel and macro-kernel approaches of operating systems with example. Write their causes.
   [2M] Dec 2024 (CSE/CET V)
3. Why do we require operating systems?
   [2M] Dec 2021 (IT V)
4. Define a trap.
   [2M] Dec 2021 (IT V)
5. Mention the objectives and functions of an operating system.
   [3M] Jun 2024 RTOS (ECE)
6. Sketch the hierarchy of storage devices.
   [3M] Jan 2023 (CSE V)
7. The response time is the major requirement of a multiuser time-sharing OS. What needs to be improved for this requirement from a system designer's viewpoint?
   [3M] Jun 2024 (AIML VI)
8. Outline the layered approach.
   [2M] Jun 2024 (CSM/AIML IV)
9. Compare desktop systems and mobile systems.
   [2M] Nov 2025 (CSE/CET V)
10. Discuss the importance of the kernel in an operating system.
    [2M] May 2025 RTOS (ECE)
11. What is an RTOS?
    [2M] May 2026 RTOS (ECE)
12. Outline the features of an RTOS.
    [3M] Jun 2024 RTOS (ECE)
13. List any four components of the Linux system.
    [2M] Dec 2021 (CSE V)

Long-answer questions (Part B)

14. The kernel of an OS masks off all interrupts during interrupt servicing. Discuss the advantages and disadvantages of such masking.
    [5M] Dec 2024 (CSE/CET V)
15. Multi-programming is inherent in multiuser and multitasking systems. Explain how.
    [4/5M] Dec 2024 (CSE/CET V); Jun 2024 (AIML VI) **Repeated x2**
16. Differentiate between multi-programming and time-sharing operating systems.
    [5M] Dec 2021 (CSE V)
17. What are the functions of an OS from the user's viewpoint and the system's viewpoint? Explain.
    [5M] Dec 2024 (CSE/CET V)

18. Explain the roles of an operating system.
[5M] Jun 2024 (CSM/AIML IV)

19. Explain the different operations performed by the operating system.
[5M] Nov 2025 (CSE/CET V)

20. Describe various activities of operating systems in connection with process management and memory management.
[5M] Dec 2021 (IT V)

21. Explain about dual-mode operation in OS with a neat block diagram.
[5M] Nov 2025 (CSE/CET V)

22. What is SPOOL? What is the benefit of spooling? Explain the difference between online and offline operation on a computer system.
[5M] Jun 2024 (AIML VI)

23. A computer system can be organized in a number of ways. Explain the categories of computer systems based on the number of general-purpose processors used.
[5M] Jan 2023 (CSE V)

24. Why is a loosely coupled system chosen over a tightly coupled system? Explain.
[5M] Jun 2024 (CSM/AIML IV)

25. Discuss symmetric multiprocessor OS considerations.
[9M] Jun 2024 RTOS (ECE)

26. Discuss, with examples, how the problem of maintaining coherence of cached data manifests itself in distributed systems.
[5M] Nov 2025 (CSE/CET V)

27. Illustrate the various categories of operating systems.
[5M] Jun 2024 (CSM/AIML IV)

28. Compare the different categories of operating systems based on their types of operations.
[4M] Jan 2023 (CSE V)

29. Write short notes on multitasking.
[5M] May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x2**

30. Describe the micro-kernel and macro-kernel approaches of operating systems with example.
[5M] Dec 2024 (CSE/CET V)

31. What is the main function of the microkernel? Explain with a neat sketch.
[5M] Jun 2024 (CSM/AIML IV)

32. What is the main advantage of the microkernel approach to OS design? How do user programs and system services interact in a microkernel architecture?
[5M] May 2025 (CSM/AIML IV)

33. What is the main advantage of the layered approach to system design? What are its disadvantages?
[4M] Jan 2023 (CSE V)

34. Explain the following operating system structures with neat sketches: (i) MS-DOS (ii) Micro-kernel (iii) iOS.
[5M] Jan 2023 (IT/AIDS V)

35. Identify advantages and disadvantages of open-source operating systems. Identify the types of people who would find each aspect an advantage or a disadvantage.
[5M] May 2025 (CSM/AIML IV)

36. How are iOS and Android similar? How are they different?
[5M] May 2025 (CSM/AIML IV)

37. Draw and explain the Android architecture.
[5M] Dec 2021 (CSE V)

38. Describe some of the challenges of designing operating systems for mobile devices compared with designing operating systems for traditional PCs.
[5M] Nov 2025 (CSE/CET V)

39. Mobile operating systems such as iOS and Android place user data and system files into two separate partitions. Aside from security, what is an advantage of that separation?
[4M] Jan 2023 (CSE V)

40. What are the primary goals of the conflict-resolution mechanism used by the Linux kernel for loading kernel modules?
[4M] Jun 2024 (AIML VI)

41. Explain the basic architecture of an RTOS with neat sketches.
[5M] May 2025 RTOS (ECE)

42. Write the differences between GPOS and RTOS.
[5M] May 2025 RTOS (ECE)

43. Discuss the importance of a real-time kernel.
[5M] May 2025 RTOS (ECE)

44. Explain RTOS with its features and applications.
[10M] May 2026 RTOS (ECE)

UNIT II - Process Management, Scheduling & IPC

Syllabus: Program vs process, process states, PCB, OS services and system calls, system programs, context switching, scheduling algorithms and criteria, Unix/Linux process management, Linux IPC, RPC.

Short-answer questions (Part A)

1. What is the purpose of the command interpreter? Why is it usually separate from the kernel?
   [2M] May 2025 (CSM/AIML IV)

2. What is the purpose of the fork() system call?
   [2M] Dec 2021 (CSE V)

3. In what circumstances is the system-call sequence fork() exec() most appropriate? When is vfork() preferable?
   [3M] Jun 2024 (AIML VI)

4. What is the state transition that occurs due to the scanf() system call in the program below? Explain.

```
int main(){
    int a;
    scanf("%d \n", &a);
    exit(0);
}
```

   [3M] Jan 2023 (IT/AIDS V)

5. What are the uses of job queues, ready queues and device queues?
   [2M] Nov 2025 (CSE/CET V)

6. Provide the diagram showing context switch from process to process.
   [3M] Jan 2023 (CSE V)

7. Context switch time is pure overhead. What factors can increase this overhead?
   [3M] Jun 2024 (AIML VI)

8. Explain the difference between preemptive and non-preemptive scheduling.
   [2M] May 2025 (CSM/AIML IV)

9. Compare high priority and low priority (scheduling).
   [2M] Jun 2024 (CSM/AIML IV)

10. What is aging? Identify the scheduling algorithm that suffers from the problem aging solves.
    [2M] Dec 2021 (CSE V)

11. Name the solution to the problem of indefinite blockage of low-priority processes.
    [2M] Dec 2021 (IT V)

12. Discuss time-sliced round-robin scheduling.
    [3M] Jun 2024 RTOS (ECE)

13. What is the use of inter-process communication?
    [2M] May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x2**

Long-answer questions (Part B)

14. Illustrate the states of a process with a diagram.
    [5M] Dec 2021 (CSE V)

15. Describe the state transition mechanisms of a process.
    [5M] Nov 2025 (CSE/CET V)

16. Discuss task (process) state transitions.
    [4M] Jun 2024 RTOS (ECE)

17. Demonstrate how a process is created and show the process state transition diagram.
    [5M] May 2025 (CSM/AIML IV)

18. What is a PCB? Outline the information it holds / describe its framework, with the application of each field in process execution.
[5M] Dec 2021 (IT V); Jun 2024 (CSM/AIML IV); Nov 2025 (CSE/CET V) **Repeated x3**

19. What are the different steps involved in process management?
[5M] May 2025 RTOS (ECE)

20. Explain context switching.
[5M] Jun 2024 (CSM/AIML IV)

21. Some computer systems provide multiple register sets. Describe what happens when a context switch occurs if the new context is already loaded into one of the register sets. What happens if the new context is in memory rather than in a register set and all the register sets are in use?
[5M] Nov 2025 (CSE/CET V)

22. A parent process needs to be terminated but has two child processes: one is waiting for I/O and the other is executing. How will you terminate the parent process? Explain.
[5M] Dec 2024 (CSE/CET V)

23. In a system, Process A is running, and Processes B and C are in a blocked state. An I/O-completion signal arrives for Process B, which has the highest priority. Describe the sequence of actions in the operating system.
[5M] Dec 2024 (CSE/CET V)

24. In a system, Process A is running and Processes B and C are blocked. Process D is ready to execute but not in the ready queue as there is no space. What will you do to accommodate D? Describe the sequence of actions in the OS.
[5M] Jun 2024 (AIML VI)

25. Justify the need for short-term, medium-term and long-term process schedulers.
[5M] Dec 2021 (CSE V)

26. Summarize / describe in detail the services provided by an operating system.
[5-10M] Dec 2021 (CSE V); Jan 2023 (IT/AIDS V); Jun 2024 RTOS (ECE); May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x5**

27. List five services provided by an operating system, and explain how each creates convenience for users. In which cases would it be impossible for user-level programs to provide these services?
[5M] Jun 2024 (AIML VI); May 2025 (CSM/AIML IV) **Repeated x2**

28. The services and functions provided by an operating system can be divided into two main categories. Briefly describe the two categories and discuss how they differ.
[4M] Jun 2024 (AIML VI)

29. Define a system call. Illustrate how system calls are used to provide operating-system services, with an example.
[5M] Dec 2021 (IT V); Jan 2023 (CSE V) **Repeated x2**

30. What is the concept of scheduling?
[5M] May 2025 RTOS (ECE)

31. Analyze the evaluation criteria for selecting a CPU-scheduling algorithm for a particular system.
[5M] Dec 2021 (IT V)

32. Discuss preemptive and non-preemptive scheduling.
[5M] Jun 2024 RTOS (ECE); May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x3**

33. Explain in detail the FCFS and SJF CPU-scheduling algorithms.
[9M] Jun 2024 RTOS (ECE)

34. Describe Round Robin scheduling.
[5M] May 2026 RTOS (ECE)

35. What is the need for multi-level queue scheduling? Explain the drawback in implementing it.
[5M] Dec 2024 (CSE/CET V)

36. Explain how process management is carried out in Linux / UNIX.
[4/5M] Dec 2021 (CSE V); Jan 2023 (CSE V) **Repeated x2**

37. Analyze the mechanisms of shared-memory and message-passing systems (IPC techniques).
[4/5M] Dec 2021 (IT V); Jan 2023 (IT/AIDS V) **Repeated x2**

38. Compare and contrast message queues and shared memory in IPC.
[5M] May 2025 (CSM/AIML IV)

39. Discuss passing of data among processes in Unix OS.
[5M] Jan 2023 (CSE V)

40. Explain IPC and network-related system calls in UNIX.
[4M] Jan 2023 (CSE V)

41. Explain the usage of sockets for inter-process communication.
[5M] Jun 2024 (AIML VI)

42. Discuss the role of mailboxes and pipes in inter-task communication.
[5M] May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x2**

43. Explain RPC.
[5M] Jun 2024 (CSM/AIML IV)

44. Classify different security challenges in IPC mechanisms and propose possible solutions.
[5M] May 2025 (CSM/AIML IV)

Numerical / problem questions

45. Quantum = 2. Draw the Gantt charts (with start and end times) using FCFS and Round Robin. Calculate average turnaround time and average waiting time.

Process	Arrival	Execution
P1	0	5
P2	1	9
P3	2	7
P4	3	2
P5	4	4

[5M] Dec 2024 (CSE/CET V)

46. (i) Draw Gantt charts using SJF and SRTF. (ii) Calculate average turnaround time. (iii) Calculate average waiting time.

Process	Burst	Arrival
P1	5	0
P2	3	1
P3	3	2
P4	1	4

[5M] Dec 2021 (CSE V)

47. Using Shortest Remaining Time First, calculate the average waiting time and average turnaround time.

Process	Arrival	CPU Burst
P1	0	3
P2	0	2
P3	2	1
P4	5	2

[5M] Jan 2023 (IT/AIDS V)

48. Use non-preemptive scheduling, deciding only on information available at decision time. (i) Average turnaround time with SJF? (ii) Compute the average turnaround time if the CPU is left idle for the first 1 unit and then SJF is used (future-knowledge scheduling). P1 and P2 wait during the idle time.

Process	Arrival	Burst
P1	0.0	8
P2	0.4	4
P3	1.0	1

[4/5M] Jun 2024 (AIML VI); Nov 2025 (CSE/CET V) **Repeated x2**

49. All arrive at time 0 in order P1..P5. (i) Draw the Gantt chart for non-preemptive priority scheduling (larger number = higher priority). (ii) Turnaround and waiting time of each process.

Process	Burst	Priority
P1	5	4
P2	3	1
P3	1	2
P4	7	5
P5	4	3

[5M] Jan 2023 (CSE V)

50. All arrive at time 0 in order P1..P5. What is the turnaround time and waiting time of each process (priority scheduling)?

Process	Burst	Priority
P1	5	4
P2	3	1
P3	1	2
P4	7	2
P5	4	3

[5M] Jun 2024 (CSM/AIML IV)

51. Preemptive, priority-based round-robin scheduling: higher number = higher priority; equal-priority processes use RR with quantum 10; a preempted process goes to the end of the queue. (i) Gantt chart. (ii) Turnaround and waiting time of each process.

Process	Priority	Burst	Arrival
P1	8	15	0
P2	3	20	0
P3	4	20	20
P4	4	20	25
P5	5	5	45
P6	5	15	55

[5M] May 2025 (CSM/AIML IV)

UNIT III - Memory Management & Secondary Storage

Syllabus: Process memory view, memory system calls (mmap, munmap, sbrk, mprotect), segmentation, paging, page faults, page replacement, page sharing, permissions, swapping, Linux memory management, disk structure and scheduling, disk management, buffering, swap space.

Short-answer questions (Part A)

1. What is paging?
   [2M] Jun 2024 (CSM/AIML IV)
2. What is the purpose of page tables in paging?
   [3M] Jan 2023 (CSE V)
3. How does paging eliminate external fragmentation?
   [2M] Dec 2024 (CSE/CET V)
4. What are the reasons for internal fragmentation and external fragmentation?
   [2M] Dec 2021 (IT V)
5. Interpret the role of the TLB in paging.
   [2M] Dec 2021 (CSE V)
6. How do you calculate effective access time?
   [2M] Dec 2021 (IT V)
7. How does an operating system handle a page fault?
   [2M] May 2025 (CSM/AIML IV)
8. What is thrashing? Explain.
   [2/3M] Dec 2021 (CSE V); Jan 2023 (IT/AIDS V) **Repeated x2**
9. Consider a logical address space of 64 pages of 1,024 words each, mapped onto a physical memory of 32 frames. How many bits are there in the logical address and the physical address?
   [2M] Nov 2025 (CSE/CET V)
10. Write the difference between synchronous and asynchronous I/O.
    [2M] Dec 2021 (CSE V)
11. Categorize the ways of accessing disk storage by computers.
    [2M] Dec 2021 (IT V)

Long-answer questions (Part B)

12. Infer the mechanism of paging and discuss the paging hardware / illustrate the paging model of logical and physical memory.
    [5M] Dec 2021 (IT V); Jun 2024 (CSM/AIML IV) **Repeated x2**
13. Discuss address translation in the two-level page table structure.
    [5M] Dec 2024 (CSE/CET V)
14. How is a logical address converted to a physical address using (a) hierarchical, (b) hashed, (c) inverted page tables?
    [5M] Jan 2023 (IT/AIDS V)
15. Explain why address-space identifiers (ASIDs) are used in TLBs.
    [5M] Nov 2025 (CSE/CET V)
16. Explain how segmentation is different from paging.
    [5M] Dec 2021 (CSE V)
17. Examine the strategies commonly used to select a free hole from the set of available holes.
    [5M] Dec 2021 (IT V)

18. Most systems allow a program to allocate more memory to its address space during execution (e.g. heap). What is required to support dynamic memory allocation in contiguous memory allocation and in paging?
[5M] May 2025 (CSM/AIML IV)

19. Discuss virtual-memory-related system calls.
[5M] May 2025 (CSM/AIML IV)

20. On a system with paging, a process cannot access memory that it does not own. Why? How could the OS allow access to additional memory? Why should it or should it not?
[5M] Nov 2025 (CSE/CET V)

21. What is the effect of allowing two entries in a page table to point to the same page frame? How could this be used to reduce the time needed to copy a large amount of memory? What effect would updating a byte on one page have on the other?
[5M] Nov 2025 (CSE/CET V)

22. Explain OS-level memory protection.
[5M] Dec 2024 (CSE/CET V); Jan 2023 (CSE V); Jun 2024 (CSM/AIML IV) **Repeated x3**

23. Explain the demand paging concept with an example.
[4M] Jan 2023 (IT/AIDS V)

24. Under what circumstances do page faults occur? Describe (with a diagram) the events / actions taken by the OS when a page fault occurs.
[4M] Jan 2023 (CSE V); Jan 2023 (IT/AIDS V) **Repeated x2**

25. Discuss the necessity for page replacement with an example.
[5M] Dec 2021 (IT V)

26. Explain the FIFO page-replacement algorithm with an example.
[5M] Jun 2024 (CSM/AIML IV)

27. With a suitable example, explain the least recently used (LRU) rule in page replacement.
[5M] May 2026 RTOS (ECE)

28. Explain the terms (a) swapping (b) overlays.
[5M] May 2025 RTOS (ECE)

29. Illustrate how memory management is performed in Linux OS.
[5M] Dec 2021 (CSE V)

30. Show the moving-head disk mechanism with a neat diagram and describe the related terminology.
[5M] Dec 2021 (IT V)

31. Explain the issues in disk management.
[4M] Jan 2023 (IT/AIDS V)

32. Interpret disk scheduling algorithms with a suitable example.
[5M] Dec 2021 (IT V)

33. Illustrate FCFS disk scheduling with a neat diagram.
[5M] Jun 2024 (CSM/AIML IV)

34. C-LOOK is an improvement over LOOK scheduling. However, it may lead to increased average response time and lower throughput. Explain how.
[4/5M] Dec 2024 (CSE/CET V); Jun 2024 (AIML VI) **Repeated x2**

Numerical / problem questions

35. A process has relocatable code of size 700 K. The relocation register is loaded with 30010 K and the limit register contains 31000 K. If the processor generates logical addresses 990 and 1020, where will they be located in physical memory?
[5M] Dec 2024 (CSE/CET V)

36. 16-bit addresses with a page size of 512 bytes. Logical address = 0000010001111101. How many bits are used for page number and offset? Compute page number and offset. What is the physical address if the frame for that page is 15?

[4M] Jun 2024 (AIML VI)

37. Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB and 125 KB (in order), how would first-fit, best-fit and worst-fit place processes of 115 KB, 500 KB, 358 KB, 200 KB and 375 KB (in order)? Which algorithm works best here?

[5M] Dec 2021 (CSE V)

38. What is segmentation? Convert logical addresses to physical addresses using the segment table below. (The specific logical addresses are not printed in the original paper.)

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580

[5M] Jan 2023 (IT/AIDS V)

39. Show the stack implementation of LRU on the string: 1 0 5 1 1 3 5 1 5 3 4 5 2 1 3 0 1 4 0 5

[5M] Dec 2024 (CSE/CET V)

40. Calculate the page faults using LRU with 3 frames: 5 0 2 1 0 3 0 2 4 3 0 3 2 1 3 0 1 5

[3M] Jun 2024 (AIML VI)

41. What is Belady's anomaly? For the string 7,2,3,1,2,5,3,4,6,7,7,1,0,5,4,6,2,3,0,1 with demand paging and three frames, how many page faults occur with Optimal and FIFO? (Nov 2025 asks Optimal only.)

[5M] Dec 2021 (CSE V); Nov 2025 (CSE/CET V) **Repeated x2**

42. Inspect the page-replacement algorithms using the reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

[5M] Dec 2021 (IT V)

43. Apply FIFO and Optimal page replacement on 1,2,3,4,1,2,3,4,2,1,3,4 with 3 frames.

[5M] Jan 2023 (CSE V)

44. Reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. How many page faults occur for LRU, FIFO and Optimal with one to seven frames (all initially empty)?

[5M] May 2025 (CSM/AIML IV)

45. Disk with 5000 cylinders (0-4999); head at 2150 moving right. Queue (FIFO): 2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681. Total head movement for SCAN and SSTF?

[5M] Dec 2021 (CSE V)

46. Disk with 5,000 cylinders (0-4,999); head at 2,150, previous request at 1,805. Queue (FIFO): 2,069; 1,212; 2,296; 2,800; 544; 1,618; 356; 1,523; 4,965; 3,681. Total distance for (i) FCFS (ii) SCAN (iii) C-SCAN.

[5M] May 2025 (CSM/AIML IV)

47. Apply FCFS and SSTF disk scheduling on 68, 78, 42, 32, 93, 35, 82 with initial head at 40. Calculate total head movement.

[5M] Jan 2023 (CSE V)

48. Disk with 201 cylinders (0-200); arm at 100; requests: 30, 85, 90, 100, 105, 110, 135, 145. Using SSTF, the request for cylinder 90 is serviced after how many requests?

[4M] Jan 2023 (IT/AIDS V)

49. A disk has 16 sectors of 1024 bytes on each track, rotating at 600 rpm. (a) How long does it take to read an entire track? (b) What volume of data can be stored if it has 10 tracks?

[5M] Jan 2023 (IT/AIDS V)

UNIT IV - Concurrency, Synchronization & Deadlocks

Syllabus: Threads (benefits, types, APIs), synchronization issues, hardware/software solutions, classical problems, deadlock conditions, RAG, prevention, avoidance, recovery.

Short-answer questions (Part A)

1. Compare process and thread.
   [2M] May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x2**
2. Distinguish between user-level threads and kernel-level threads.
   [2M] Dec 2021 (CSE V)
3. List the benefits of threads.
   [2M] Jun 2024 (CSM/AIML IV)
4. What is a race condition?
   [2M] Dec 2021 (CSE V)
5. What is the difference between Dekker's solution and Peterson's solution for two-process synchronization?
   [2M] Dec 2024 (CSE/CET V)
6. What are semaphores, and how are they used in synchronization?
   [2M] May 2025 (CSM/AIML IV)
7. List the features of a mutex semaphore.
   [2M] May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x2**
8. Summarize / write about the necessary conditions for a deadlock to occur.
   [2/3M] Dec 2021 (IT V); Jan 2023 (IT/AIDS V) **Repeated x2**
9. Identify the need for the RAG (resource-allocation graph) in deadlock.
   [2M] Dec 2021 (CSE V)
10. Is it possible to have a deadlock involving only one single-threaded process? State the reason.
    [2M] Nov 2025 (CSE/CET V)

Long-answer questions (Part B)

11. Define thread and discuss the multithread concept.
    [5M] May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x2**
12. Compare and contrast process and thread with respect to any five parameters.
    [5M] Dec 2021 (CSE V)
13. Explain multithreading models with suitable examples / write a short note on the multithreading model.
    [4/5M] Jan 2023 (IT/AIDS V); May 2025 RTOS (ECE); May 2026 RTOS (ECE) **Repeated x3**
14. Explain the implementation of kernel threads.
    [5M] Dec 2024 (CSE/CET V)
15. Would you classify Linux threads as user-level threads or kernel-level threads? Support your answer.
    [5M] Jan 2023 (CSE V)
16. Can a multithreaded solution using multiple user-level threads achieve better performance on a multiprocessor system than on a single-processor system? Explain.
    [4M] Jan 2023 (CSE V)
17. Analyze the importance of threading libraries.
    [5M] May 2025 RTOS (ECE)
18. Interpret / describe Peterson's solution to the critical-section problem (two-process synchronization).
    [4/5M] Dec 2021 (IT V); Jan 2023 (IT/AIDS V) **Repeated x2**
19. What is the meaning of busy waiting? Can busy waiting be avoided altogether? Explain.
    [4M] Jan 2023 (CSE V)

20. Explain why spinlocks are not appropriate for single-processor systems yet are often used in multiprocessor systems.
[5M] Nov 2025 (CSE/CET V)

21. What is a semaphore? Explain its initial implementation.
[5M] Dec 2024 (CSE/CET V)

22. Explain the wait and signal operations used in semaphores. What are the applications of semaphores?
[5M] Dec 2021 (CSE V)

23. Show that if the wait() and signal() semaphore operations are not executed atomically, mutual exclusion may be violated.
[5M] Nov 2025 (CSE/CET V)

24. Describe the operations on a mutex.
[5M] May 2026 RTOS (ECE)

25. Demonstrate any one classic problem of synchronization in detail.
[5M] Dec 2021 (IT V)

26. Analyze the structure of the consumer process (producer-consumer problem).
[5M] Jun 2024 (CSM/AIML IV)

27. Develop a solution to the Readers-Writers problem using semaphores / how is it handled using synchronization techniques?
[5M] Dec 2021 (CSE V); May 2025 (CSM/AIML IV) **Repeated x2**

28. Explain the dining-philosophers problem.
[5M] Jun 2024 (CSM/AIML IV)

29. Explain the deadlock and starvation problems in the dining-philosophers problem.
[5M] Dec 2024 (CSE/CET V)

30. A working couple has three babies. The babies cannot be left unattended: one of the couple must be at home. Write a solution for synchronizing the parents with the babies.
[5M] Jun 2024 (AIML VI)

31. Define deadlock and explain the necessary conditions for deadlock.
[5M] May 2025 RTOS (ECE)

32. Describe how deadlock can occur in a system. Discuss methods to avoid or handle deadlock.
[5M] May 2026 RTOS (ECE)

33. Prove that the circular-wait condition may be produced if all the other three necessary conditions of a deadlock are present.
[5M] Dec 2024 (CSE/CET V)

34. Interpret how the hold-and-wait condition contributes to deadlock. Suggest a way to prevent it.
[5M] May 2025 (CSM/AIML IV)

35. Illustrate the resource-allocation graph with a deadlock.
[5M] Jun 2024 (CSM/AIML IV)

36. Describe how deadlock is possible with the bounded-buffer problem.
[5M] Nov 2025 (CSE/CET V)

37. A multithreaded application uses only reader-writer locks for synchronization. Applying the four necessary conditions, is deadlock still possible if multiple reader-writer locks are used?
[5M] Nov 2025 (CSE/CET V)

38. Explain the Banker's algorithm with illustrative steps for deadlock avoidance.
[5M] Jan 2023 (CSE V); Jun 2024 (CSM/AIML IV) **Repeated x2**

39. How do you use deadlock avoidance with multiple instances of resource types? Explain with one illustrative example.

[4M] Jun 2024 (AIML VI)

40. Compare the circular-wait scheme with deadlock-avoidance schemes (like the Banker's algorithm) with respect to runtime overhead and system throughput.

[5M] May 2025 (CSM/AIML IV)

41. Why do deadlocks occur? Illustrate the deadlock detection and recovery mechanism.

[5M] Dec 2021 (CSE V); Dec 2021 (IT V) **Repeated x2**

Numerical / problem questions

42. For the code segment below: (i) how many unique processes are created? (ii) how many unique threads are created?

```
pid_t pid;
pid = fork();
if (pid == 0) {   /* child process */
    fork();
    thread_create(...);
}
fork();
```

[5M] May 2025 (CSM/AIML IV)

43. Two processes use the construct below; wants1 and wants2 are shared and initialized to false. Does it satisfy mutual exclusion, progress and bounded waiting? Justify.

```
/* P1 */                        /* P2 */
while (true) {                 while (true) {
  wants1 = true;                 wants2 = true;
  while (wants2 == true);        while (wants1 == true);
  /* Critical Section */         /* Critical Section */
  wants1 = false;                wants2 = false;
}                                }
/* Remainder section */        /* Remainder section */
```

[5M] Jan 2023 (IT/AIDS V)

44. Identify the Need matrix and determine whether the system is in a safe state by demonstrating an order in which processes may complete. Available = (3, 3, 2, 1).

	Allocation A B C D	Max A B C D
P0	2 0 0 1	4 2 1 2
P1	3 1 2 1	5 2 5 2
P2	2 1 0 3	2 3 1 6
P3	1 3 1 2	1 4 2 4
P4	1 4 3 2	3 6 6 5

[5M] Dec 2021 (CSE V)

45. Using the Banker's algorithm, determine whether the system is in a safe state. Available = (0, 3, 0, 1).

	Allocation A B C D	Max A B C D
P0	3 0 1 4	5 1 1 7
P1	2 2 1 0	3 2 1 1
P2	3 1 2 1	3 3 2 1
P3	0 5 1 0	4 6 1 2
P4	4 2 1 2	6 3 2 5

[4M] Jan 2023 (IT/AIDS V)

UNIT V - File Systems & OS Security

Syllabus: File concepts, types, allocation and organization, file system calls, metadata, directory structure, caching, case study; threats, basic security mechanisms, malware taxonomy, viruses, worms, rootkits, logging, auditing, recovery.

Short-answer questions (Part A)

1. List the (basic) attributes of a file.
   [2M] Dec 2021 (CSE V); Dec 2021 (IT V) **Repeated x2**
2. Explain file system mounting.
   [3M] Jan 2023 (IT/AIDS V)
3. Why is a two-level directory better than a single-level directory?
   [2M] Jun 2024 (CSM/AIML IV)
4. What is journaling?
   [3M] Jan 2023 (CSE V)
5. What are the advantages and disadvantages of providing mandatory locks instead of advisory locks whose use is left to users' discretion?
   [2M] Nov 2025 (CSE/CET V)
6. Differentiate threat and attack.
   [2M] Dec 2021 (IT V)
7. Differentiate between vulnerability, threat and attack.
   [3M] Jun 2024 (AIML VI)
8. What do you mean by password vulnerabilities?
   [3M] Jan 2023 (CSE V)
9. List the security attacks that cannot be prevented by encryption.
   [2M] Dec 2024 (CSE/CET V)
10. What is a privilege-escalation attack? How does it affect OS security?
    [2M] May 2025 (CSM/AIML IV)
11. What are capability-based systems?
    [2M] Dec 2021 (IT V)
12. List the goals of protection.
    [3M] Jan 2023 (IT/AIDS V)

Long-answer questions (Part B)

13. Explain the different file types.
    [5M] Jun 2024 (CSM/AIML IV)
14. Describe the various file organization methods with illustrative examples.
    [5M] Nov 2025 (CSE/CET V)
15. Explain the various file allocation methods (allocating disk space for files); mention the format of the file allocation table in each method.
    [5M] Dec 2021 (CSE V); Dec 2021 (IT V); Jan 2023 (IT/AIDS V) **Repeated x3**
16. Explain different methods for free-space management.
    [4M] Jan 2023 (CSE V)
17. Explain the mapping of a logical file system to physical devices.
    [5M] Jun 2024 (CSM/AIML IV)
18. Describe the file-system actions at close() with an illustrative example.
    [5M] Dec 2024 (CSE/CET V)

19. Explain the most common schemes for defining the logical structure of a directory.
[5M] Dec 2021 (IT V); Jun 2024 (AIML VI) **Repeated x2**

20. Analyze the two-level directory structure.
[5M] Jun 2024 (CSM/AIML IV)

21. Explain file directory structure and caching optimizations.
[5M] Nov 2025 (CSE/CET V)

22. What is file-system caching? Explain its benefits in file access.
[5M] May 2025 (CSM/AIML IV)

23. What are the main advantages of journaling file systems like ext3, ext4 and NTFS?
[5M] May 2025 (CSM/AIML IV)

24. A file named data is frequently accessed by users. Compare the advantages and drawbacks of (a) setting up links from every user's home directory to data, and (b) copying data into every user's home directory.
[5M] Dec 2024 (CSE/CET V)

25. Discuss some standard security attacks in operating systems.
[5M] Jun 2024 (AIML VI)

26. Analyze the most common system and network threats.
[6M] Dec 2021 (IT V)

27. Describe different types of program threats with suitable examples.
[4M] Jan 2023 (IT/AIDS V)

28. Explain the malware taxonomy with a detailed description.
[5M] Dec 2024 (CSE/CET V)

29. Illustrate viruses and worms.
[5M] Jun 2024 (CSM/AIML IV)

30. Illustrate how Trojans and backdoors compromise OS security.
[5M] May 2025 (CSM/AIML IV)

31. Discuss a means by which managers of Internet-connected systems could limit or eliminate the damage done by worms. What are the drawbacks of the change you suggest?
[4M] Jan 2023 (CSE V)

32. Buffer-overflow attacks can be avoided by better programming methodology or special hardware support. Discuss these solutions.
[5M] Jan 2023 (CSE V); Nov 2025 (CSE/CET V) **Repeated x2**

33. Explain how Data Execution Prevention (DEP) prevents memory-based attacks.
[5M] May 2025 (CSM/AIML IV)

34. Discuss security levels in operating systems.
[4M] Jun 2024 (AIML VI)

35. Demonstrate / explain the role of cryptography as a security tool.
[4/5M] Dec 2021 (IT V); Jan 2023 (IT/AIDS V) **Repeated x2**

36. List the goals and principles of protection in a modern computer system.
[4M] Dec 2021 (IT V)

37. How can the access matrix be implemented effectively? Explain the methods used.
[5M] Dec 2021 (IT V); Jan 2023 (IT/AIDS V) **Repeated x2**

38. Describe the operations performed on the access matrix.
[5M] Jan 2023 (IT/AIDS V)

39. Explain the control access right on the access matrix.
[5M] Jun 2024 (AIML VI)

40. Discuss the strengths and weaknesses of implementing an access matrix using capabilities associated with domains.

[5M] Nov 2025 (CSE/CET V)

41. Explain and distinguish between logging and auditing.

[4M] Jun 2024 (AIML VI)

Left out as out of syllabus

These appeared in the papers but aren't covered by the five units above (mostly from the ECE Real-Time OS papers).

- RAID levels (Dec 2021 (CSE V), Jun 2024 (AIML VI))
- Kernel I/O structure (Dec 2021 (CSE V))
- Vi editor and its commands (Jun 2024 (AIML VI))
- UNIX consistency semantics for remote file systems (Jun 2024 (AIML VI))
- Booting process (May 2026 RTOS (ECE))
- Rate-monotonic and earliest-deadline-first scheduling; timing-constraint classification (Jun 2024 RTOS (ECE), May 2026 RTOS (ECE))
- Priority inversion; dynamic priority levels in RTOS (Jun 2024 RTOS (ECE))
- Commercial RTOS comparison (QNX, uC/OS-II, RT-Linux) (Jun 2024 RTOS (ECE))
- VxWorks-specific: features, taskCreate(), task creation/activation/deletion and safety, binary/counting semaphores in VxWorks (Jun 2024 RTOS (ECE), May 2025 RTOS (ECE), May 2026 RTOS (ECE))
- Memory-mapped I/O in RTOS; real-time garbage collection; block memory management (May 2025 RTOS (ECE), May 2026 RTOS (ECE))
- Linux development tools/setup, debugger goals, embedded Linux architecture, GNU cross-platform (Jun 2024 RTOS (ECE))